BINSTD toy-yield citation provenance audit

Validation audit, 22 October 2026

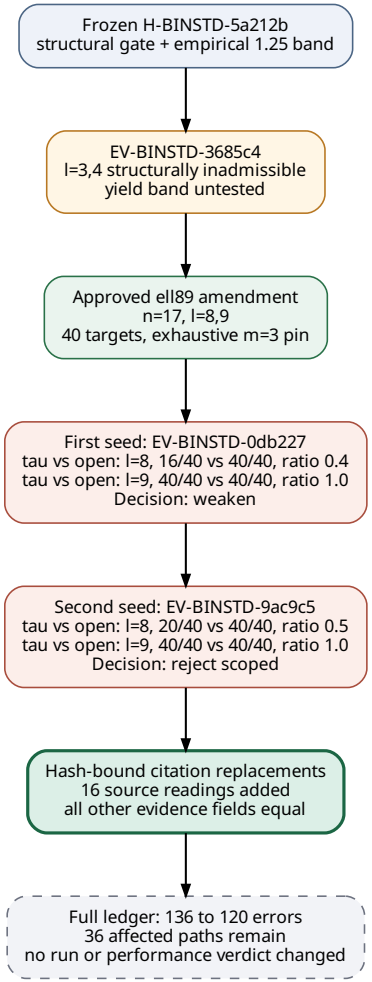
Question. Do the two archived BINSTD yield evidence records have source-bound internal citations, and do their aggregate ratios agree with the frozen run results?

Source chain. The original $n = 17$, $\ell \in \{3, 4\}$ panel found no admissible invariant dimensions and left the yield band untested. An approved additive amendment moved the panel to $\ell \in \{8, 9\}$. A second approved amendment repeated that panel under a new Stage-0 seed. Both use 40 targets and the toy exhaustive $m = 3$ factor-base-sum pin.

Panel	ℓ	τ hits	open hits	R_τ / R_{open}
First seed	8	16/40	40/40	0.4
First seed	9	40/40	40/40	1.0
Second seed	8	20/40	40/40	0.5
Second seed	9	40/40	40/40	1.0

Verification. The archived Stage-1 matrix hashes match their run-result digests. A focused checker recomputes four ratios from the published aggregate hits and confirms that removing only the 16 new source-reading notes makes both replacements structurally identical to their frozen records. On parent commit `5e499ba86dfa`, the full ledger validator falls from 136 errors in 38 paths to 120 in 36: exactly 16 citation errors removed, none added. It still exits 1.

Boundary. All open arms hit 40/40, so the 1.25 ratio cannot be met at those observed open-arm rates. This audit does not replay per-target solver work, certify target sampling, or prove empirical yield in Lean. The first review's preliminary weaken and the second review's toy-scope rejection are unchanged. No break, asymptotic or deployed-security conclusion follows.


Frozen H-BINSTD-5a212b
structural gate + empirical 1.25 band
EV-BINSTD-3685c4
l=3,4 structurally inadmissible
yield band untested
Approved ell89 amendment
n=17, l=8,9
40 targets, exhaustive m=3 pin
First seed: EV-BINSTD-0db227
tau vs open: l=8, 16/40 vs 40/40, ratio 0.4
tau vs open: l=9, 40/40 vs 40/40, ratio 1.0
Decision: weaken
Second seed: EV-BINSTD-9ac9c5
tau vs open: l=8, 20/40 vs 40/40, ratio 0.5
tau vs open: l=9, 40/40 vs 40/40, ratio 1.0
Decision: reject scoped
Hash-bound citation replacements
16 source readings added
all other evidence fields equal
Full ledger: 136 to 120 errors
36 affected paths remain
no run or performance verdict changed